IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-493

1. Topic (12 words max)

AI-Driven Optimization System for Yield, Quality, Disease, and Water Monitoring in Hydroponic Lettuce.

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Hydroponic lettuce cultivation is rapidly expanding in Sri Lanka as a sustainable solution to challenges like limited arable land, unpredictable weather, and the need for efficient resource use. However, accurate growth monitoring and yield prediction remain difficult, as current systems rely heavily on sensor data, such as pH, electrical conductivity, temperature, and humidity- which, while precise, often overlook subtle indicators of plant stress, disease, or nutrient deficiency that experienced growers can detect. [1] This lack of integration between sensor analytics and human intuition leads to delayed interventions and inconsistent yields, particularly in small and medium-scale farms without access to advanced automation. Farmers in these setups depend on hands-on experience, yet their observations are rarely digitized or systematically used in decision-making. Local climate variations, like monsoons or heatwaves, further complicate the accuracy of predictions. [2] To bridge this gap, the research proposes a smart, grower-friendly system that fuses real-time sensor data with structured human input.

Hydroponic lettuce farms in Sri Lanka still rely mainly on manual inspection to identify plant diseases and nutrient deficiencies, leading to delays and significant yield loss. While recent Sri Lankan research has explored IoT-based hydroponic monitoring, these systems focus on environmental factors and do not capture visible leaf symptoms critical for early intervention. [3] Advances in deep learning, such as YOLO-based image analysis, now enable real-time detection of defective leaves and visible deficiencies in hydroponic lettuce. This research proposes an automated IoT camera system using the YOLOv8 model to detect disease symptoms and nutrient deficiencies as soon as they appear. [4] The system assigns health scores and provides actionable dashboard suggestions, reducing manual effort and enabling data-driven crop management in local hydroponic setups.

Lettuce is highly perishable, and maintaining its freshness during storage is challenging. Traditional manual inspections are inefficient and lack real-time accuracy. Environmental factors such as temperature, humidity and gases like ethylene accelerate spoilage, but continuous monitoring is rarely implemented. [5] This research focuses on reliable, real-time data acquisition from sensors to capture these environmental parameters inside lettuce storage rooms. Smarter monitoring and prediction systems are essential to maintain quality and reduce waste. [6] Research in food technology highlights that multiple factors (temperature, humidity, gas emissions, etc.) influence spoilage. Ethylene gas, for example, plays a critical role in ripening and can dramatically shorten freshness. [7] However, human operators cannot continuously track such parameters or accurately extrapolate the remaining shelf-life. This is where Artificial Intelligence (AI) and Internet of Things (IoT) sensors can intervene.

In hydroponic lettuce farming, water quality directly affects plant health, yield, and root development. However, farmers often lack real-time insight into key parameters such as pH, nutrient strength (EC), temperature, dissolved oxygen, and algae presence. In tropical regions like Sri Lanka, warm and humid conditions accelerate algae growth and oxygen depletion, leading to root damage and reduced crop quality. [8] Existing systems rarely integrate algae detection or early oxygen warnings. This research addresses the problem by developing a smart, low-cost water quality monitoring system that continuously checks critical parameters, detects algae risk, and monitors dissolved oxygen. The system provides real-time alerts and recommendations to prevent water-related issues, ensuring healthier crops and reduced waste.

References

[1] J. R. Gowtham Rajendiran, "Smart Aeroponic Farming System: Using IoT with LCGM-Boost Regression Model for Monitoring and Predicting Lettuce Crop Yield," 2023.

[2] J. R. Gowtham Rajendiran, "Lettuce Crop Yield Prediction Analysis using Random Forest Regression Machine Learning Model in Aeroponics System," 2023.

[3] N. A. B. D. R. K. Sewmina, "Understanding the Need for IoT-based NFT Hydroponic Systems: A case study in Colombo, Sri Lanka," 2023.

[4] X. J. e. al., "Hydroponic lettuce defective leaves identification based on improved YOLOv5s.," 2023.

[5] U. B. R. A. ,. A. M. B. ,. M. H. Ekta Sonwani, "An Artificial Intelligence Approach Toward Food Spoilage Detection and Analysis," 2022.

[6] †. M. ,. A. 2. K. 1. a. T. Flora Vitalis 1, "Aquaphotomics Monitoring of Lettuce Freshness during Cold Storage," 2023.

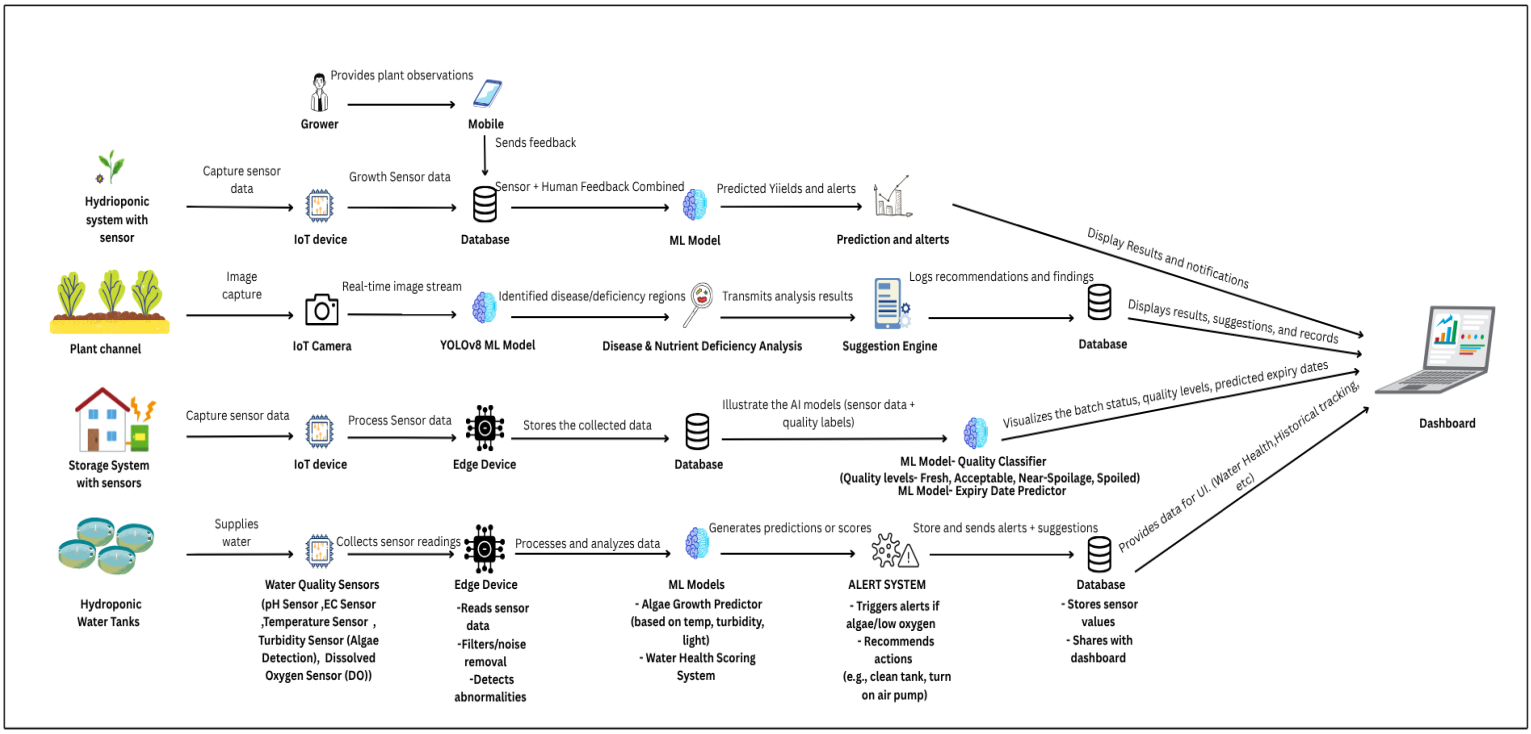
[7] Q. W. a, "Insights of freshness phenotype detection for postharvest fruit and vegetables," 2025.

[8] J. I. S. Simon Prananta Barus, "Temperature and Humidity Monitoring in Hydroponic Cultivation Based on Internet of Things: Dataset Development for Smart Agriculture," 2025.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a hybrid growth monitoring and yield prediction system that combines automated sensor data with structured grower feedback, forming a human-in-the-loop AI platform for hydroponic lettuce farming. The system continuously collects key environmental parameters- such as pH, electrical conductivity, temperature, and humidity, via sensors installed in the hydroponic setup, while simultaneously allowing growers to input qualitative observations, such as signs of leaf discoloration, unusual textures, or other indicators of plant health, through a user-friendly mobile or web application. These two complementary data streams, quantitative sensor readings and qualitative human insights are integrated into a central database, generating a rich, multi-modal dataset for each crop cycle. A machine learning model trained on this data predicts expected yields and issues early warnings if anomalies or deviations from expected growth patterns are detected. Results are visualized through an intuitive dashboard that displays predictions, trends, and alerts, enabling growers to make timely, informed decisions. This integrated approach leverages both technological precision and human expertise to deliver more accurate and actionable yield forecasts.

A dashboard presents all predictions and alerts to the end-user. Additionally, an IoT camera system powered by a YOLOv8 model enables real-time detection of plant diseases and nutrient deficiencies. For water quality, sensors capture parameters like pH, EC, turbidity, and dissolved oxygen to predict algae risk and issue alerts. This research also focuses on designing and deploying an IoT sensor network to measure temperature, humidity, CO_2, and ethylene levels in the storage environment. These readings are processed by edge computing devices and fed into AI models that classify lettuce quality (e.g., Fresh, Good, Borderline, Spoiling) and predict remaining shelf-life using regression or time-series models.


Provides plant observations
Grower
Mobile
Sends feedback
Hydrioponic system with sensor
Capture sensor data
IoT device
Growth Sensor data
Database
Sensor + Human Feedback Combined
ML Model
Predicted Yiields and alerts
Prediction and alterts
Display Results and notifications
Plant channel
Image capture
IoT Camera
Real-time image stream
YOLOv8 ML Model
Identified disease/deficiency regions
Disease & Nutrient Deficiency Analysis
Transmits analysis results
Suggestion Engine
Logs recommendations and findings
Database
Displays results, suggestions, and records
Storage System with sensors
Capture sensor data
IoT device
Process Sensor data
Edge Device
Stores the collected data
Database
Illustrate the AI models (sensor data + quality labels)
ML Model- Quality Classifier
(Quality levels- Fresh, Acceptable, Near-Spoilage, Spoiled)
ML Model- Expiry Date Predictor
Visualizes the batch status, quality levels, predicted expiry dates
Dashboard
Hydroponic Water Tanks
Supplies water
Water Quality Sensors
(pH Sensor ,EC Sensor ,Temperature Sensor , Turbidity Sensor (Algae Detection), Dissolved Oxygen Sensor (DO))
Collects sensor readings
Edge Device
-Reads sensor data
-Filters/noise removal
-Detects abnormalities
Processes and analyzes data
ML Models
- Algae Growth Predictor (based on temp, turbidity, light)
- Water Health Scoring System
Generates predictions or scores
ALERT SYSTEM
- Triggers alerts if algae/low oxygen
- Recommends actions
(e.g., clean tank, turn on air pump)
Store and sends alerts + suggestions
Database
- Stores sensor values
- Shares with dashboard
Provides data for UI. (Water Health,Historical tracking, etc)

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Implementing this hybrid monitoring and prediction system requires combined expertise in software engineering and agricultural science. Technically, the system demands skills in IoT integration, mobile/web development, database management, and machine learning, particularly in time-series forecasting, data fusion, and user interface design. Reliable sensor calibration and data preprocessing are crucial for accurate readings of pH, EC, temperature, and humidity. On the agricultural side, knowledge of optimal growing conditions for hydroponic lettuce, common stress symptoms (like tip burn or yellowing), and the effects of Sri Lanka's climate is essential. Structuring consistent, actionable grower feedback also requires basic user training and thoughtful data encoding. The system relies on continuous sensor data (ideally sampled hourly), regular human feedback (daily or weekly), and final yield data (harvested weight or count) to train and validate the machine learning model. While historical datasets combining sensor and grower data are ideal, pilot deployments can provide initial training data, and open-source datasets can further support model development. Collaboration between engineers, agricultural experts, and growers is key to ensuring the system's accuracy and real-world effectiveness.

Developing this solution requires an interdisciplinary blend of domain expertise across agriculture, sensor technology, and artificial intelligence. On the technology side, IoT and sensor system expertise is needed to design and implement the hardware network. This includes selecting appropriate sensors (temperature/humidity probes, CO_2 and ethylene gas sensors, etc.), calibrating them, and ensuring reliable data transmission (possibly via WiFi or Bluetooth from the storage site to the cloud). AI expertise is at the core of developing the predictive model. Skills in machine learning (particularly time-series analysis or predictive modeling) are required to create algorithms that can learn from environmental data patterns and correlate them with lettuce spoilage outcomes.

8. Objectives and Novelty

Main Objective

This project aims to develop an AI-powered, IoT-based system to improve quality monitoring, yield prediction, and disease detection in hydroponic lettuce farming. It integrates four components: sensor-based shelf-life prediction, a hybrid model combining grower input with sensor data for yield forecasting, real-time disease and deficiency detection using IoT cameras and YOLOv8, and a water quality monitoring system with algae and oxygen alerts. Together, these solutions enhance decision-making, reduce waste, and support sustainable lettuce cultivation in Sri Lanka.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22312358 IMALKA T A D D	Automated real-time detection and management of lettuce diseases and nutrient deficiencies using IoT camera and edge AI.	Deploy IoT cameras in hydroponic lettuce systems to capture plant images continuously. Integrate YOLOv8 deep learning model on an edge device for real-time image analysis. Detect and classify visible diseases and nutrient deficiencies on lettuce leaves. Assign health scores based on severity and type of detected issues. Generate and deliver actionable care recommendations through a dashboard.	Simultaneously detects lettuce diseases and nutrient deficiencies in real time using continuous IoT camera monitoring and on-device YOLOv8 analysis, and uniquely provides quantitative health scores with automated, actionable care recommendations for Sri Lankan hydroponic growers.

		Store historical health data and recommendations for growers' decision support.	
IT21390050 Sachintha PWE	To design and implement a real-time water quality monitoring system for hydroponic lettuce farming that detects pH, EC, temperature, algae presence, and oxygen levels to maintain optimal root-zone conditions.	Integrate and calibrate water quality sensors (pH, EC, temperature, turbidity, DO). Develop microcontroller-based system to process sensor data in real time. Implement alert mechanism for low oxygen or high algae risk. Train ML model for algae prediction and water health scoring. Visualize sensor data and alerts on shared dashboard.	This system uniquely integrates turbidity-based algae detection and dissolved oxygen monitoring into hydroponic farming, which are often overlooked. With machine learning for algae prediction and water health scoring, it offers proactive, low-cost water quality management, enhancing root-zone health and addressing critical challenges in tropical environments like Sri Lanka.
IT22332158 Thennakoon G H	Develop and validate a machine learning model to predict lettuce quality status (fresh/good/ borderline/spoiling) and estimate remaining shelf-life from sensor inputs.	Collect and preprocess multi-sensor data. Train and fine-tune ML algorithms for classification & regression. Evaluate model accuracy against actual shelf-life. Deploy the trained model within the end-to-end system.	Utilizes multi-sensor data fusion (temperature, humidity, CO_2, ethylene/VOC levels) for shelf-life prediction- an innovative approach for lettuce, as conventional methods or prior models typically consider only temperature or time. This datadriven AI model offers a novel precision in predicting lettuce spoilage before it is visible.

		Visualize sensor data and alerts on the dashboard.	
IT22330178 FERNANDO A N B	To develop and validate a hybrid system that combines real-time sensor data and structured grower feedback for more accurate growth monitoring and yield prediction in hydroponic lettuce farming.	Design and implement a data collection platform for both sensor streams and grower feedback. Integrate and preprocess sensor and human input data into a unified database. Develop and train a machine learning model that uses both data types for yield prediction and anomaly detection. Visualize predictions and alerts through a user-friendly dashboard. Evaluate system performance against traditional, sensor-only approaches.	human-in-the-loop AI system for hydroponic lettuce yield prediction and growth monitoring, systematically fusing quantitative sensor data with structured qualitative grower feedback to enhance prediction accuracy and early problem detection.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22312358 IMALKA T A D D	This component applies to computer vision and embedded AI principles by integrating IoT cameras and YOLOv8 deep learning models for automated lettuce health analysis in hydroponic systems. The design uses modular, edge-based inference to enable real-time detection of diseases and nutrient deficiencies directly on plant leaves. Results are processed and translated into actionable recommendations following software engineering best practices for reliability and scalability. The solution leverages agricultural domain expertise for dataset creation and effective grower guidance through a user-friendly dashboard.
IT21390050 Sachintha PWE	This component applies to core principles of software engineering through the development of a real-time IoT-based system for monitoring water quality in hydroponic lettuce farming. It involves sensor integration (pH, EC, temperature, turbidity, DO), embedded system programming, and the design of a modular, scalable architecture using microcontrollers. The system processes and transmits data to a central database and dashboard, incorporating alert mechanisms and visualization tools. It also supports optional AI modules for algae prediction and water health scoring, demonstrating full-stack development and system interoperability. The component reflects specialization in software engineering by combining hardware-software integration, data flow design, and user-oriented interface development to ensure reliable, maintainable, and efficient operation in a real-world agricultural context.
IT22332158 Thennakoon G H	The component involves designing and implementing an AI-powered predictive model for lettuce shelf-life, aligning closely with software engineering and artificial intelligence specialization. The AI component uses machine learning models such as Random Forest, XGB, SVM, and LSTM trained on sensor data (temperature, humidity, CO_2, ethylene) to classify lettuce quality and predict expiry dates. By processing time-series environmental data, these models identify patterns indicative of freshness decline and spoilage. This approach leverages data analytics and supervised learning to transform raw sensor inputs into actionable predictions, aligning with software engineering and AI specialization by applying advanced algorithms to real-world agricultural quality monitoring without relying on image data.
IT22330178 FERNANDO A N B	This component integrates core principles of software engineering by combining IoT-based sensor data collection, mobile/web application development for human feedback, database management, and machine learning for predictive analytics. The solution leverages specialized knowledge in data fusion, time-series analysis, and user interface design to create an intelligent, user-centric system. By enabling seamless interaction between hardware, software, and human expertise, it demonstrates

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	practical application of software engineering methodologies to solve real-world agricultural challenges in hydroponic yield prediction.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Samanthi	Siriwardana	
Co-Supervisor	Ms.	Mihiri	Samaraweera	27/Jul/25
External Supervisor	Mr.	Dinuka	Thennakoon	

Summary of external supervisor's (if any) experience and expertise

Head of Plantation Manager of Asia Plantation & Export (PVT) LTD. As a skilled Assistant Superintendent with 5 years' experience and 1 year experience as a Senior Assistant Superintendent in plantation.

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.